BWEM Pilot Study – Scenario 1: Alfalfa Irrigation Efficiency Comparison

This report documents Scenario 1 of the BioLoop Water Efficiency Model (BWEM) pilot simulations. The scenario evaluates a 10-acre alfalfa farm with 20 dairy cattle under semi-arid climatic conditions. Two irrigation systems were compared: Flood irrigation (60% efficiency) and Drip irrigation (90% efficiency). Colorado / semi-arid reference ET

1. Methodology

Crop evapotranspiration (ETc) was calculated using the FAO-56 crop evapotranspiration method:

ETc = ETo × Kc

Where ETo represents reference evapotranspiration (mm/day) and Kc represents the crop coefficient. The simulations used ETo = 4.0 mm/day and Kc = 0.85 for alfalfa. Livestock water demand was estimated based on CSU Extension livestock water consumption guidelines (~35 gallons/day per dairy cow).

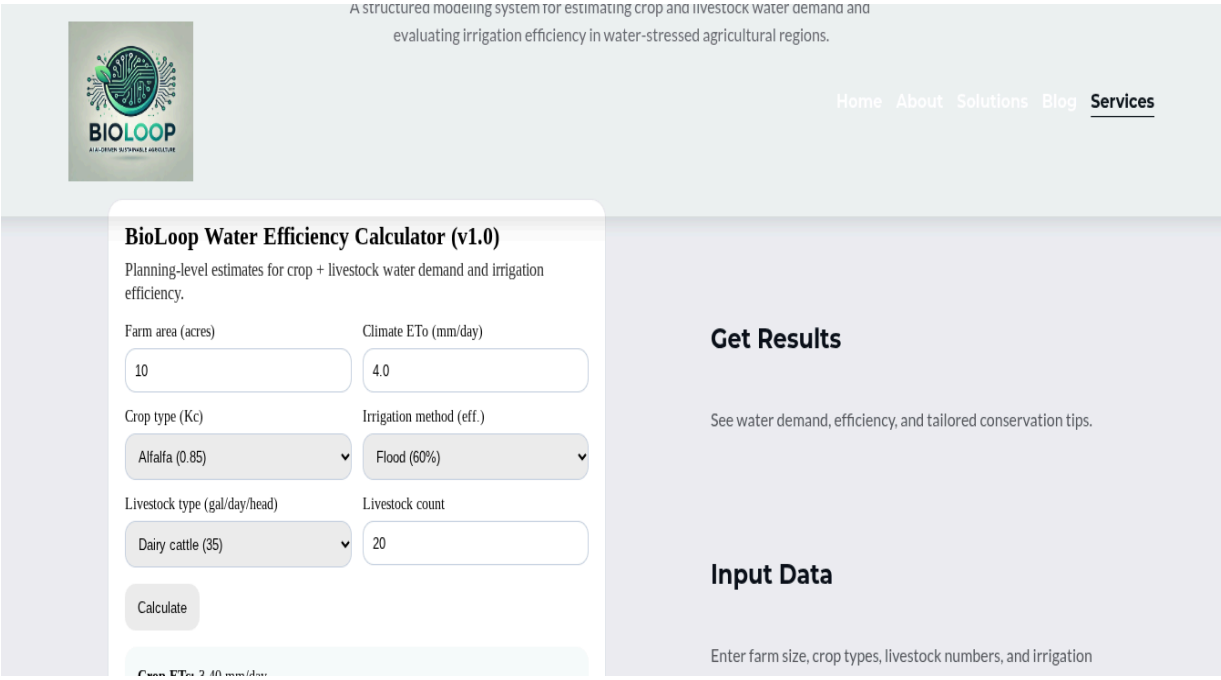
2. Simulation Results

Scenario	Farm Size	Crop	Livestock	Total Demand	Applied Water	Water Loss
Flood (60%)	10 acres	Alfalfa	20 dairy	37,048 gal/day	61,280 gal/day	24,232 gal/day
Drip (90%)	10 acres	Alfalfa	20 dairy	37,048 gal/day	41,087 gal/day	4,039 gal/day

Drip irrigation reduces estimated water loss from 24,232 to 4,039 gal/day (≈83% reduction).

3. Screenshots from BWEM Simulation

Figure 1. Scenario 1 input and output (first irrigation case)



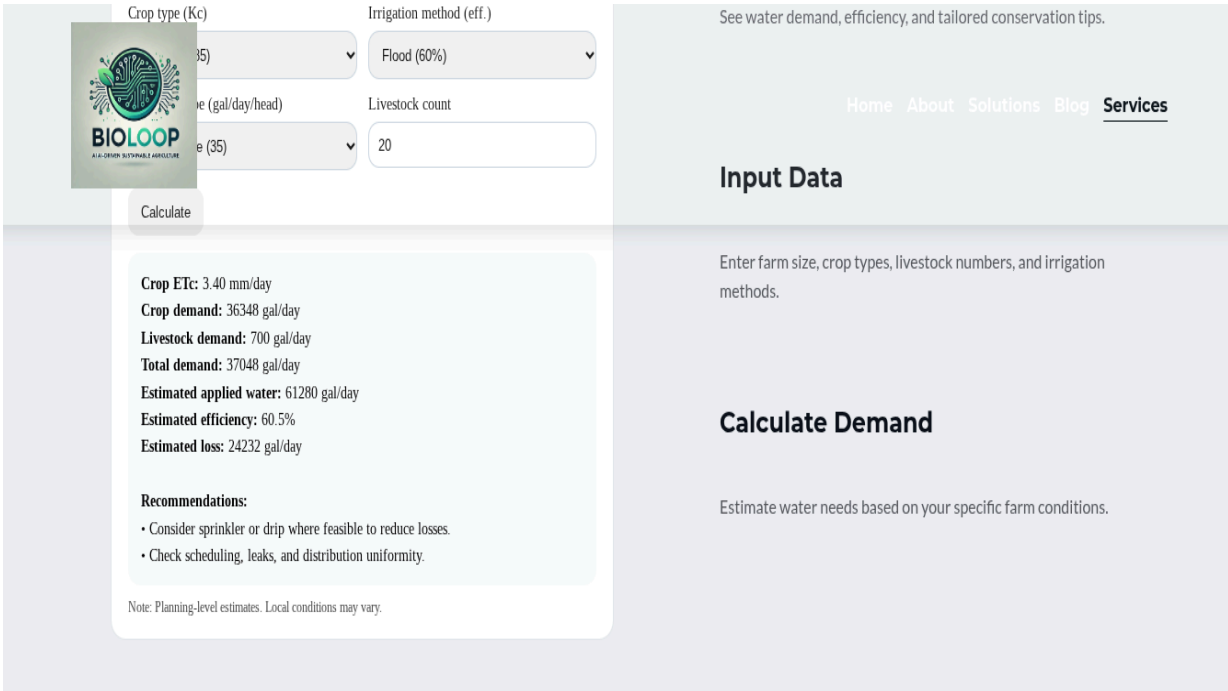


Figure 2. Scenario 1 input and output (second irrigation case)

evaluating irrigation efficiency in water-stressed agricultural regions.

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BioLoop Water Efficiency Calculator (v1.0)

Planning-level estimates for crop + livestock water demand and irrigation efficiency.

Farm area (acres): 10

Climate ETo (mm/day): 4.0

Crop type (Kc): Alfalfa (0.85)

Irrigation method (eff.): Drip (90%)

Livestock type (gal/day/head): Dairy cattle (35)

Livestock count: 20

Calculate

Crop ETc: 3.40 mm/day

Get Results

See water demand, efficiency, and tailored conservation tips.

Input Data

Enter farm size, crop types, livestock numbers, and irrigation methods.

See water demand, efficiency, and tailored conservation tips.

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Irrigation method (eff.): Drip (90%)

Livestock count: 20

Calculate

Crop ETc: 3.40 mm/day
Crop demand: 36348 gal/day
Livestock demand: 700 gal/day
Total demand: 37048 gal/day
Estimated applied water: 41087 gal/day
Estimated efficiency: 90.2%
Estimated loss: 4039 gal/day

Recommendations:
• Outputs indicate reasonable efficiency; monitor seasonally and adjust as needed.

Note: Planning-level estimates. Local conditions may vary.

Input Data

Enter farm size, crop types, livestock numbers, and irrigation methods.

Calculate Demand

Estimate water needs based on your specific farm conditions.

4. References

Allen, R.G., Pereira, L.S., Raes, D., & Smith, M. (1998). *Crop Evapotranspiration – Guidelines for Computing Crop Water Requirements.* FAO Irrigation and Drainage Paper No. 56. Food and Agriculture Organization of the United Nations.

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Colorado State University Extension. *Calculating Water Requirements for Cattle.* https://goldenplains.extension.colostate.edu/wp-content/uploads/sites/56/2021/06/PR_CalculatingWaterforCattle.pdf

Crop Coefficients for Use in Irrigation Scheduling. https://irrigationtoolbox.com/ReferenceDocuments/Extension/BCExtension/577100-5.pdf

E1 Pilot Study Enhancement — Alfalfa Scenario

Practical Interpretation

The results demonstrate how BWEM can be used to compare irrigation strategies and estimate water demand for alfalfa under different irrigation methods. Differences between irrigation methods can inform decisions regarding water allocation, irrigation efficiency, and resource optimization in semi-arid agricultural settings.

Use Case Scenario

This model can support pre-season agricultural planning by allowing users to estimate crop water requirements for alfalfa and compare irrigation options. It may assist farmers, advisors, and planners in evaluating different irrigation approaches before implementation.

Limitations

This pilot is based on simplified assumptions and is intended for planning-level use only. It does not account for detailed field variability, real-time environmental conditions, or water quality constraints. The model is designed as a decision-support tool rather than a precise operational system.